



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGIES

SPRING 2026

FINAL EXAMINATION

COURSE TITLE:	SOFTWARE ARCHITECTURE
COURSE CODE:	SEN3244
INSTRUCTOR:	Engr. TEKOH PALMA
DATE:	1 st Week after Spring Written Exams 2026
DURATION:	8 Weeks

Project specification:

Project Presentation Period: 1st Week after Spring Written Exams 2026

Instructions:

You are required to **design and implement an innovative Architectural solution for software project** of your choice, either individually or in a team (max 4 members). Your project should solve a real-world problem and must incorporate all the sections below. Marks are allocated based on the completeness, originality, technical depth, and correct use of architectural principles and tools.

1. Infrastructure Setup (15 Marks)

Acquire, set up and configure the necessary **VPS** infrastructure to deploy your application:

- Provision cloud/on-premise resources.
- Install all necessary technologies to support your project development.
- Configure networking, firewall rules, reverse proxies, etc.
- Use Docker, Kubernetes or any other relevant environment tools.

Deliverables:

- Infrastructure diagram

- Terraform/Bash scripts (optional but appreciated)
- Screenshots and links showing infrastructure deployment

2. Application of Scrum (5 Marks)

Demonstrate the application of Agile practices in your project:

- Identify roles: Product Owner, Scrum Master, Team
- Maintain a product backlog and sprint backlog
- Include a burndown chart for at least 2 sprints

Deliverables:

- Document showing your scrum roles, sprint planning, and retrospectives
- Screenshots from tools like Jira, Trello, or GitHub Projects (if applicable)

3. CI/CD Pipeline with Jenkins (10 Marks)

Implement a CI/CD pipeline using Jenkins:

- Automate build, test, and deployment stages
- Integrate your source control (e.g., GitHub/GitLab)

Deliverables:

- Jenkinsfile
- Screenshots or demo video of Jenkins pipeline runs
- Explanation of the pipeline stages

4. Continuous Monitoring with Prometheus and Grafana (2.5 Marks)

Set up platform and application monitoring:

- Export Metrics from your Platform and your Application
- Visualize and alert using Prometheus and Grafana

Deliverables:

- Dashboard screenshots
- Explanation of key metrics being monitored

5. Infrastructure as Code with Ansible (2.5 Marks)

Provision and configure part of your infrastructure using Ansible:

- Include at least 2 playbooks (e.g., install packages, start services)

Deliverables:

- Ansible playbooks
- Execution logs/screenshots

6. Robust Testing of Application (10 Marks)

Implement testing at different levels:

- Unit tests, Integration tests, and optionally E2E tests
- Achieve minimum 80% code coverage

Deliverables:

- Test results and coverage report
- Sample test cases and automation scripts

7. Containerization and Orchestration with Kubernetes (15 Marks)

Deploy your application using Kubernetes:

- Containerize with Docker
- Deploy with Kubernetes manifests or Helm charts
- Implement scaling, rolling updates, and service discovery

Deliverables:

- Dockerfiles
- Kubernetes YAMLs or Helm chart
- Screenshots or recordings of Kubernetes dashboard or CLI output

8. Architecture Structures and Characteristics (20 Marks)

Demonstrate architectural thinking:

- Identify and justify architecture style (e.g., microservices, layered, event-driven)

- Show key architectural structures (component, deployment, module views)
- Discuss trade-offs and quality attributes (e.g., performance, scalability, security).
- Pros and Cons of chosen Architecture

Deliverables:

- Architecture document with diagrams (UML encouraged)
- A report showing all the steps of the Architectural design process as discussed during the course.

9. Project Innovation (10 Marks)

Show originality and creativity in your project:

- Use of emerging tech (e.g., AI, blockchain, AR)
- Unique business logic or compelling user experience

Deliverables:

- Description of innovation aspect
- Demo video or screenshots highlighting innovation

10. Project Documentation (15 Marks)

Provide comprehensive documentation:

- README (setup, usage, contribution guidelines) on Github Repo
- API documentation (Swagger/Postman preferred)
- User manual or onboarding guide
- Project Report Template

Deliverables:

- Repository with complete documentation
- Link to hosted Swagger or Postman collection (if API-based)

NB: Project Reports should follow the template below deliverables such as images and descriptions should be included in the report.

PROJECT REPORT TEMPLATE

Course Code/Course Title

Group Number

Project Topic

Link to GitHub Repository

Group Leader

Group Information

<i>SN</i>	<i>Member's Name</i>	<i>Registration Number</i>	<i>Team Role</i>
<i>1</i>			
<i>2</i>			
<i>3</i>			
<i>4</i>			

Note:

- *The following format at minimum must be used for the Project Presentation (This is just a guide but you are not limited to it, you are allowed to expand it further)*

CHAPTER ONE: INTRODUCTION

- ❖ *General Introduction*
- ❖ *Aim and Objectives*
- ❖ *Problem Statement*

CHAPTER TWO: LITERATURE REVIEW

- ❖ *Software Development Methodologies*
- ❖ *Comparison between different Software Development Methodologies*

- ❖ *Reason for the choice of Scrum Methodology*
- ❖ *General review of related concepts with respect to your chosen project*
- ❖ *Review of related literature with respect to your chosen project*

CHAPTER THREE: METHODOLOGY AND MATERIALS

- ❖ *Research Methodology*
- ❖ *System Requirements (Functional and Non-Functional)*
- ❖ *System Design*
 - *Architecture of your system (HLD)*
 - *UML Diagrams*
- ❖ *Application of scrum*
 - *Team organization*
 - *Workflow management*
 - *Conflict Resolution*
 - *Challenges encounter and how you overcame them*
- ❖ *Scrum Artifacts (Product backlog and Sprint backlog)*
- ❖ *A Test Case document*
- ❖ *Proposed Algorithms*
- ❖ *Materials and technologies used (briefly list the name of the technology or material and its role in realizing your system)*

CHAPTER THREE: RESULTS AND DISCUSSIONS

- ❖ *Screenshots of various application scenarios*
- ❖ *Screenshots of various API Request/Response*
- ❖ *Etc..*

CHAPTER FOUR: RECOMMENDATIONS AND CONCLUSION

In not more than 3 paragraphs summarize what your team has been able to achieve, the difficulties you encountered and recommendation for further studies

Submission Guidelines

Students are expected to submit the following artifacts.

- A printed Report (Word or Latex) which follows strictly the Report template provide above.
- Include a 7-minute video walkthrough of your project
- PowerPoint slide (not more than 20slides)
- .ZIP Project folder containing all the deliverables (source code, configuration files etc).

Important to Know

- The link to the submission form will be provided by the lecturer
- Submission deadline: **The 1st Week After Spring written Exams**
- Late submission penalty: **Will be Evaluated on 60%**